



ST BENEDICT'S

SUBJECT	Physical Science: Paper 1	DATE	16 November 2020
GRADE	11	MARKS	
EXAMINER NAME	Mr McCoy	MODERATOR	Mr du Toit
CLASS	MARKING GUIDELINES	DURATION	3 hours

LOW ORDER 50%

HIGH ORDER 50%

QUESTION 1 - GR

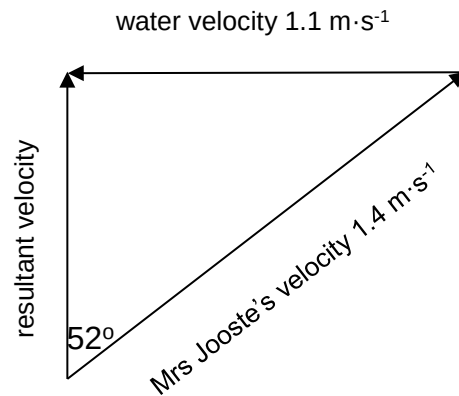
20 MARKS

- 1.1 B ✓✓
 - 1.2 D ✓✓
 - 1.3 C ✓✓
 - 1.4 D ✓✓
 - 1.5 A ✓✓
 - 1.6 A ✓✓
 - 1.7 B ✓✓
 - 1.8 D ✓✓
 - 1.9 B ✓✓
 - 1.10 D ✓✓
-

QUESTION 2 - MRS JOOSTE

25 MARKS

2.1.1



✓✓✓ each arrow and label
✓ angle

(4)

2.1.2 Define resultant vector as the single vector ✓ which has the same effect as the original vectors acting together ✓

(2)

2.1.3 $1,4^2 \checkmark - 1,1^2 \checkmark = x^2$ subs

$x = 0,87 \text{ (0,866)} \text{ m} \cdot \text{s}^{-1} \checkmark$ at 90° to the river bank directly across. ✓

(4)

2.1.4

$$v = \frac{\Delta x}{\Delta t} \checkmark$$

$$\text{c.o. } 0,866 \checkmark = 89 \checkmark / \text{t}$$

$t = 102,8 \text{ s} \checkmark$ therefore yes she makes it (only award this mark if they compare to the 5 mins using same unit) ✓

(5)

2.2

2.2.1 the rate of change of position **or** the rate of displacement **or** the rate of change of displacement ✓✓

(2)

2.2.2 They are travelling at the same velocity (speed) ✓

(1)

2.2.3

$$a = \text{gradient} \checkmark = \frac{20 - 4 \checkmark}{8} = 2 \text{ m} \cdot \text{s}^{-2}$$

$$\checkmark \text{ Car A: } s = 4t + \frac{1}{2}(2)t^2$$

$$\checkmark \text{ Car B: } s = 20t$$

$$s_A = s_B$$

$$4t + t^2 = 20t \checkmark$$

$$t = 16 \text{ s} \checkmark \text{ (or } 20t = 4t + t^2 \checkmark \checkmark \text{)}$$

$$s_A = (20)(16) = 320 \text{ m} \checkmark$$

✓✓ $2 \text{ m} \cdot \text{s}^{-2}$ for car A
✓ subs for car A
✓ subs for car B
✓ link that $s_A = s_B$
✓ answer for time
✓ answer for s_A

(7)

QUESTION 3 - MRS JOOSTE**18 MARKS**

3.1.1 $x = \text{area} = \frac{1}{2} b \times h + \frac{1}{2} b \times h$ ✓ (x = area has to be stated for the formula mark)

$$= \frac{1}{2} 8 \times 5 \text{ ✓} + \frac{1}{2} 2 \times 5 \text{ ✓}$$

$$= 25 \text{ m ✓}$$

(4)

3.1.2 $v_f = v_i + at$ ✓

$$0 = 5 \text{ ✓} + 2a \text{ ✓}$$

$$a = -2,5 \text{ m} \cdot \text{s}^{-2}$$

$$a = 2,5 \text{ m} \cdot \text{s}^{-2} \text{ backwards ✓}$$

OR $a = \frac{\Delta v}{\Delta t}$ ✓

$$= \frac{0 - 5 \text{ ✓}}{10 - 8 \text{ ✓}}$$

$$= -2,5 \text{ m} \cdot \text{s}^{-2}$$

$$a = 2,5 \text{ m} \cdot \text{s}^{-2} \text{ backwards ✓}$$

(4)

3.3 Furthest point away is 25 m.

Acceleration is $2,5 \text{ m} \cdot \text{s}^{-2}$ at 10 s.

$$x = v_i t + \frac{1}{2} at^2 \text{ ✓}$$

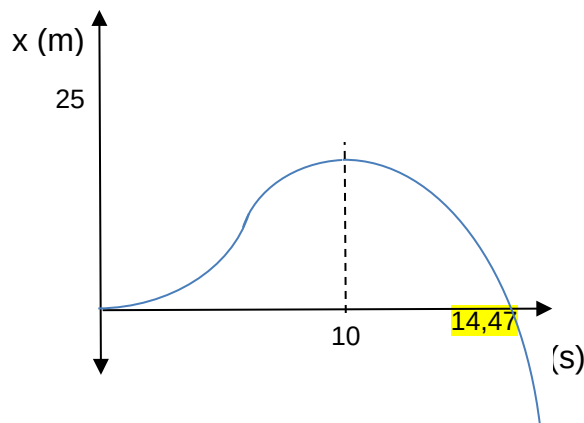
$$25 \text{ ✓} = 0 + \frac{1}{2} (2,5 \text{ ✓}) t^2 \text{ (subs)}$$

$$t = 4,47 \text{ s ✓}$$

$$t = 10 + 4,47 = 14,47 \text{ s ✓}$$

(5)

3.4



- ✓ Up curve from zero
- ✓ Down curve to horizontal
- ✓ Down curve past zero
- ✓ y axis value
- ✓ x axis values

(5)

QUESTION 4 – MR MCCOY

17 MARKS

4.1 Increasing velocity (downwards) ✓✓ OR constant/ uniform acceleration (downwards) ✓✓ (2)

4.2 $v = u + gt$ ✓

$$55 = 15 + 9,8t$$
 ✓

$$t = 4,08 \text{ s}$$
 ✓

(3)

4.3 $v^2 = u^2 + 2gs$ ✓

$$55^2 = 15^2 + 2(9,8)s$$
 ✓

$$s = 142,86 \text{ m}$$
 ✓

(3)

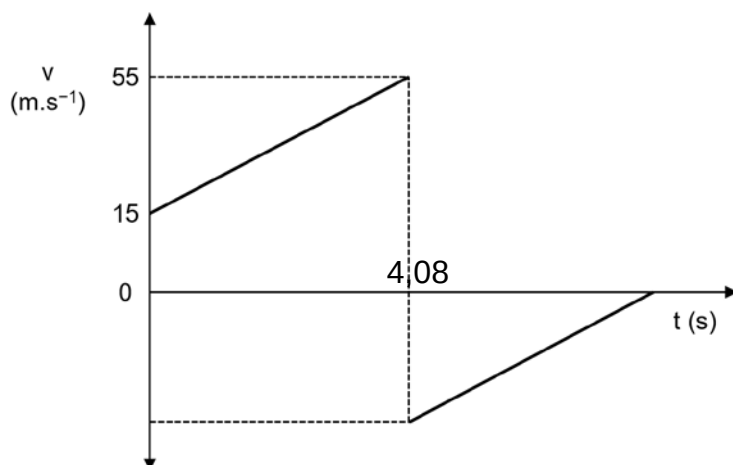
4.4 $142,86 \text{ (3.3 coe)} - 100 \text{ (minus 100)} = 42,86 \text{ m}$ ✓

(2)

4.5 The gradient at C is **zero** ✓✓ AON

(2)

4.6 (5)



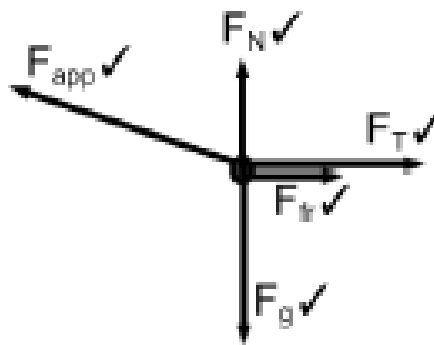
- ✓ Starting at 15 m.s^{-1}
- ✓ positive gradient to 55 m.s^{-1}
- ✓ vertical line at $4,08 \text{ s}$
- ✓ parallel gradient
- ✓ Ending at zero v

QUESTION 5 – MR MCCOY

32 MARKS

5.1.1 All labels written in full.

(5)



5.1.2 The gravitational force of the 5 kg mass, pulling up in the Earth. (3)

5.1.3a $F_{\text{vert}} = F \sin \theta = 60 \sin 10 = 10,42 \text{ N}$ (2)

5.1.3b $F_{\text{hor}} = F \cos \theta = 60 \cos 10 = 59,09 \text{ N}$ (2)

5.1.4 as the perpendicular force exerted by a surface on an object in contact with it (2)

5.1.5 $F_N = F_g - F_{\text{vert}} = (5 \times 9,8) - 10,42 = 38,58 \text{ N}$ (3)

5.1.6 as the force that opposes the motion of an object and acts parallel to the surface with which the object is in contact (2)

5.1.7 $F_f = \mu F_N = 0,5 \times 38,58 = 19,29 \text{ N}$ (2)

5.1.8 For 5 kg block: $(59,09 - T - 19,29) = 5a \therefore 39,8 - T = 5a$

For 2 kg block: $T - (2 \times 9,8) = 2a \therefore T - 19,6 = 2a$

Solve simultaneously: $39,8 - (2a + 19,6) = 5a \therefore 20,2 = 7a \therefore a = 2,89 \text{ m.s}^{-2}$

$T = 2(2,89) + 19,6 = 25,37 \text{ N}$ (6)

5.2 In an accident the external force is on the car

The inertia of the passenger keeps them moving forward,

Newton's First Law

The seat belt prevents the passenger from moving forward

(Hence, reducing injury as the passenger)

will not crash into the dashboard or windscreen.

(5)

QUESTION 6- MR DU TOIT**35 MARKS**

- 6.1.1 as the product of the displacement ✓ and the component of the force parallel to the displacement ✓ (2)
- 6.1.2 **Ambiguous question 6 marks were awarded to all candidates.**
 $W_{\text{net}} = \Delta E_k$ ✓
 $W_a + W_f = \frac{1}{2} m (v^2 - u^2)$
 $(300)(2) \checkmark - F_f(2 \checkmark) = \frac{1}{2} (200 \checkmark)(3^2 - 5^2 \checkmark)$
 $600 - 2F_f = -1600$
 $F_f = 1100 \text{ N left} \checkmark$ (6)
- 6.1.3a $\sin 30 = 1,5/x$
 $x = 3 \text{ m} \checkmark \checkmark$ (2)
- 6.1.3b work done by a net force ✓ on an object is equal to the change in the kinetic energy of the object ✓ (2)
- 6.1.3c $E_k = \frac{1}{2} m (v^2 - u^2) \checkmark$
 $= \frac{1}{2} (200) \checkmark (0^2 - 3^2 \checkmark)$ must be in the correct order.
 $= -900 \text{ J}$
 $\therefore 900 \text{ J} \checkmark$ lost (4)
- 6.1.3d $W_{\text{net}} = \Delta E_k$
 $W_a + W_f + \Delta E_p \checkmark = -900 \text{ (c.o.e)}$
 $F_a(3) \checkmark - (400)(3) \checkmark - (200)(9.8)(\sin 30 \times 3) \checkmark = -900$
 $F_a = 1080 \text{ N} \checkmark$ (6)
- 6.1.3e $W_a = F_a \times s \checkmark$
 $= 1080 \text{ c.o.e} \times (3) \checkmark$
 $= 3240 \text{ J} \checkmark$ (3)
- 6.2.1 $\Sigma W = W_a + W_f + \Delta E_p \checkmark$
 $1294,3 \checkmark = 2000 + W_f - 343 \checkmark$
 $W_f = 362,7 \text{ J} \checkmark$ (4)
- 6.2.2 $P = W/t \checkmark$
 $= 2000/1,16 \checkmark$
 $= 1724,14 \text{ W} \checkmark$ (3)
- 6.2.3 $\% \text{ efficiency} = \frac{\text{power/energy}_{\text{out}}}{\text{power/energy}_{\text{in}}} \times 100 \checkmark$
 $= \frac{2000 - 362,7}{2000} \checkmark \times 100$
 $81,87 \% \text{ efficient} \checkmark$ (3)
-

QUESTION 7 – MR NTOMBELA**17 MARKS**

7.1 a material's opposition to the flow of electric current ✓✓ (AON) (2)

7.2 $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{10} = \frac{4}{15}$ ✓

$R_p = 3,75 \Omega$

$R_{ext} = 3,75 + 4$ ✓ (adding 4) $= 7,75 \Omega$ ✓

(3)

7.3 $V_4 \Omega = I \times R$
 $= 3 \times 4$
 $= 12 \text{ V}$

$I = V/R$

$= (23,25 - 12) \text{ ✓} / 6 \text{ ✓}$

$= 1,88 \text{ (1,875) A} \text{ ✓}$

Ratio method

6:10 ✓

1,125:1,1875 ✓

(3)

7.4 $P = I^2 \times R \text{ ✓}$ OR $P = V^2/R$
 $= 3^2 \times 4 \text{ ✓}$ $= 12^2 / 4$
 $= 36 \text{ W} \text{ ✓}$

(3)

7.5 External resistance (R_{ext}) increases ✓
therefore current (I) decreases ✓ (relationship between I and R_{ext})

$R = V/I$, voltage is directly proportional to current ✓
the reading would decrease. ✓

(4)

7.6 Emf - the total energy supplied per coulomb of charge ✓ by the cell ✓

(2)

7.7 23,25 V ✓

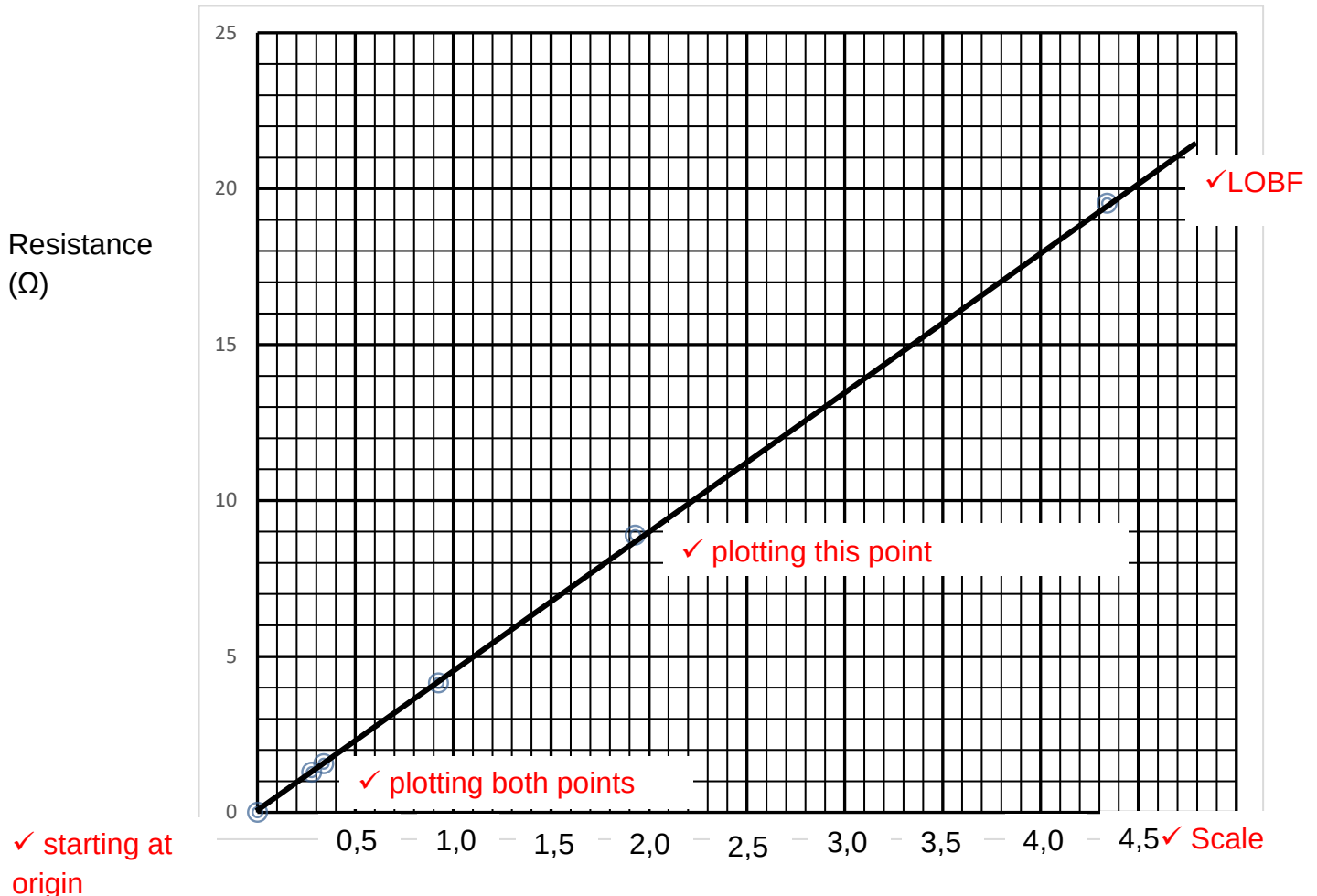
Reads the potential difference of the battery ✓

(2)

QUESTION 8 – MR NTOMBELA

22 MARKS

- 8.1 effect of gauge ✓ of a metallic conducting wire ✓ (2)
- 8.2 Temperature ✓ and length. ✓ (2)
- 8.3 1,93 ✓✓ (2)
- 8.4 Heading: graph of Resistance vs $\frac{1}{\text{gauge}^2}$. ✓ Heading (7)



$$\frac{1}{\text{gauge}^2} (\times 10^6 \text{ m}^{-2}) \quad \checkmark \text{ label and unit}$$

- 8.5 The resistance of a wire is inversely proportional ✓ to the square of the gauge of a wire. ✓ (Must be in this order, minus one if not in words) (2)
- 8.6 Gradient formula ✓
 Substitution, must be 2 ✓
 = (mark range) 5×10^{-6} ✓
 Unit = $\Omega \cdot \text{m}^2$ ✓ (4)
- 8.7 Gradient = R/A^{-1}
 $\therefore R \cdot A$ ✓
 $R = \rho \frac{L}{A}, \therefore R \cdot A = \rho L$ ✓
 Gradient = ρL ✓ (3)

QUESTION 9 – MR DU TOIT

12 MARKS

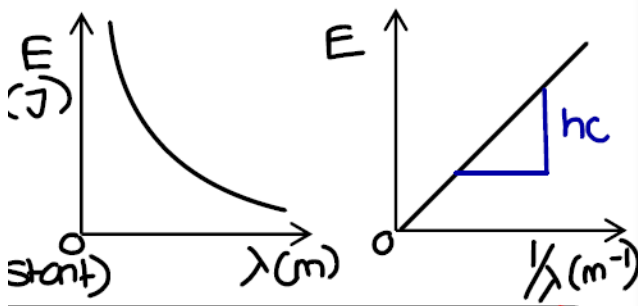
9.1

Energy is absorbed ✓
 ground state ✓ into an excited state ✓
 When dropping release photon ✓
 Different transitions equate to different wavelengths of light. ✓

(5)

9.2 Sketch either graph below. Shape ONLY ✓✓

(2)



9.3 Formula ✓
 substituting in constants hc ✓
 Conversion ✓
 Answer ✓
 Conclusion ✓

$$E_{\text{red photon}} = hc / \lambda$$

$$= 6.6 \times 10^{-34} \times 3 \times 10^8 / (658 \times 10^{-9})$$

$$= 3 \times 10^{-19} \text{ J}$$

$$\Delta E_A = (-1.52 - (-3.4)) \times 1.6 \times 10^{-19} = 3 \times 10^{-19} \text{ J}$$

$$\Delta E_B \neq 3 \times 10^{-19} \text{ J}$$

∴ **A** results in the red photon.

(5)

